Paleomagnetism of Rocks from the White Mountain Plutonic-Volcanic Series in New Hampshire and Vermont[1]

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Samples were collected at 17 sites in gabbros, monzonites, and diorites of the White Mountain plutonic-volcanic series in New Hampshire and Vermont. No useful information was obtained from the diorites. The main mass of the White Mountain series has been dated at 180 m.y. After partial demagnetization by alternating fields to remove unstable components of magnetization, the directions of magnetization became very well grouped; both normal and reversed directions are present. The mean direction and parameters for 12 sites are $D = 358\frac{1}{2}°$, $I = 59°$, $R = 11.9$, and $K = 110$, yielding a paleomagnetic pole at $85\frac{1}{2}°$N and $124\frac{1}{2}°$E with $p = 4\frac{1}{2}°$, $m = 6°$. Samples were taken from two sites in the gabbro of the Mount Ascutney intrusive in southeastern Vermont. This instrusive is younger than the rocks of the White Mountain plutonic-volcanic series and has been dated by K-Ar at 125 to 135 m.y. These sites gave paleomagnetic pole positions which are in agreement with other Upper Jurassic-Cretaceous pole positions from North America.

INTRODUCTION

The White Mountain plutonic-volcanic series in New Hampshire and eastern Vermont is composed of a series of instrusives with some associated extrusives. The instrusives range in composition from gabbro to granite. In any particular suite the basic rocks are the oldest. Acidic intrusives form the greatest mass of the exposed rocks of the series. The rocks have been intruded into schists, gneisses, and intrusives of mainly Paleozoic age. No major tectonic phenomena have taken place in the area since the emplacement of the White Mountain plutonic-volcanic series. Figure 1 is a generalized map of the New Hampshire–Vermont area showing the site localities and the main instrusives of the White Mountain series [*Chapman*, 1954; *Quinn*, 1944; *Freedman*, 1950; *Modell*, 1936; *Chapman and Chapman*, 1940]. Six cores per site were taken at seventeen sites in the White Mountain series using a small diamond drill. At each site the cores were oriented using both a Brunton compass and a sun compass. The cores with a diameter of 2.54 cm were cut into specimens 2.34 cm in length.

The Conway granite of the White Mountain series gives concordant K-Ar, Rb-Sr, and U-Pb (isotopic) ages around 180 m.y. [*Tilton et al.*, 1957; *Hurley et al.*, 1960], which indicates an Early Jurassic age on the *Kulp* [1961] time scale. *Faul et al.* [1963] report K-Ar ages of 125 and 135 m.y. for biotites from the Mount Ascutney, Vermont, ring-dike complex, which would place this intrusive in the Upper Jurassic or Lower Cretaceous. This latter instrusive has generally been thought to be related to the White Mountain series.

THE NATURAL REMANENT MAGNETIZATION

The directions and intensities of the NRM were measured with a 5-cps spinner magnetometer which is in principle much like the 100-cps spinner magnetometer described by *Phillips* [1965]. The directions of magnetization are measured to an accuracy of 1° and the intensity to better than 5%. The magnetometer was calibrated for intensity by passing a known current through a coil with an accurately determined radius and a known number of turns. The intensity calibration was checked against specimens in which the moment had been measured with other instruments. The sensitivity of the instrument at the present time is about 1×10^{-7} emu/cm^3 per specimen of 10 cm^3. Spurious effects from anisotropic rocks are minimized by reducing the earth's field to a low

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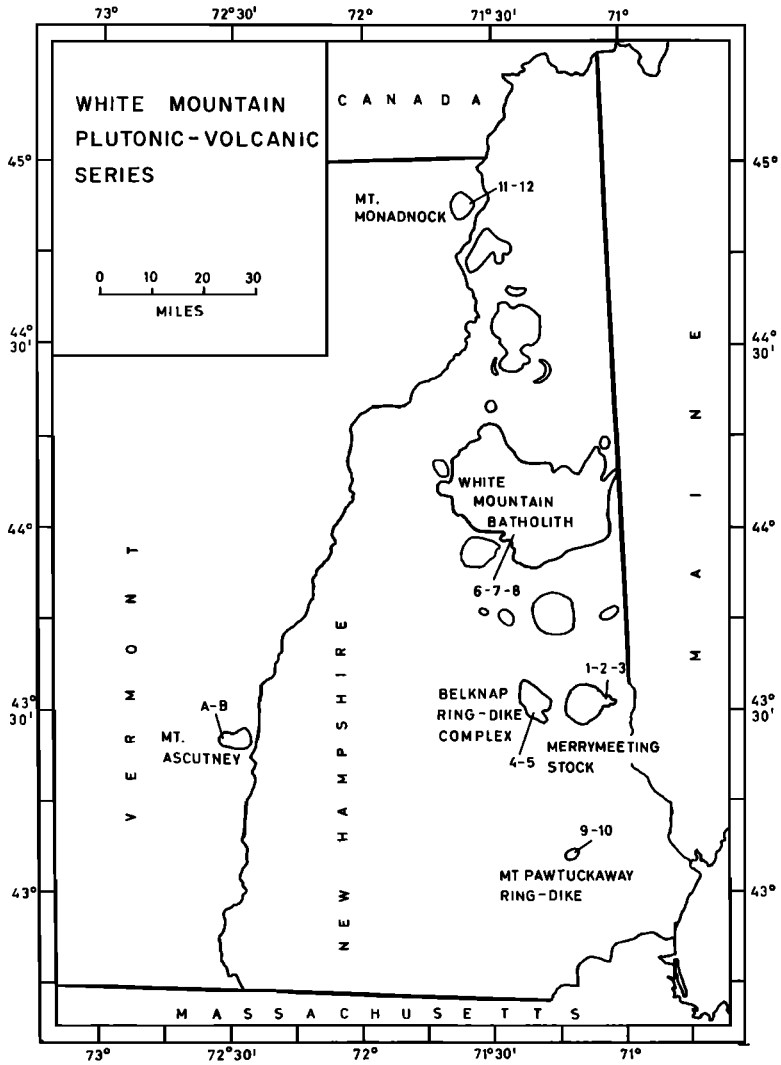


Fig. 1. Geological sketch map of New Hampshire and Vermont showing sampling sites.

level in the space where the rocks are spinning. The directions of NRM at most sites are widely scattered, showing positive inclinations (north-seeking directions pointing downward). Sites 6, 7, 8, and 9, however, show reversed directions with negative inclinations (south-seeking directions pointing downward). In general, the NRM directions are close to the present direction of the earth's magnetic field. The parameters of the NRM at each site are presented in Table 1.

Magnetic Parameters

The volume susceptibility of each core was measured on a commercial susceptibility bridge to an absolute accuracy of 5%. The mean susceptibility of each site is given in Table 1. The variation in susceptibilities of the *gabbros* is higher than that of the monzonites. The magnetic moments are much more variable than the susceptibilities within any one site, and again the more basic rocks have higher values.

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TABLE 1.

Site No.	Rock Type	Site Position: N Lati-tude	Site Position: W Longi-tude	N	E	Magnetic Properties: K	Magnetic Properties: J	M_0: D, deg	M_0: I, deg	M_0: R	M_0: k	B_{ac}	M_{ac}: D, deg	M_{ac}: I, deg	M_{ac}: R	M_{ac}: k	M_{ac}: α_{95}	Site Pole: Lati-tude	Site Pole: Longi-tude
1	Gabbro	43°35′	71° 5′	12	0	3 68	0 98	320	+57	11.060	13	300	200	−52½	11.881	93	4½	71½°N	46°E
2	Gabbro	43°35′	71° 5′	12	0	3.62	2.01	330½	+76	11.141	13	500	184	−55	11.768	47	6½	81½°N	86½°E
3	Gabbro	43°35′	71° 5′	11	0	2.18	1.69	289	+83	10 095	10	500	191	−57	10.612	26	9	79½°N	51°E
4	Monzonite	43°30′	71°20′	6	5	1 86	0.94	5	+70½	5.965	146	150	0	+65	5.908	54	9	86½°N	71½°W
5	Monzonite	43°30′	71°20′	11	0	1 06	0.41	354½	+54	7.857	3	150	2	+59	10.846	65	5½	86°N	86°E
6	Gabbro	43°58′	71°28′	12	0	3 78	7.28	11½	−59½	11.287	15	250	168	−57½	11.675	34	7½	79°N	170°E
7	Basalt Dike	43°58′	71°28′	12	0	6.62	2.68	64	−54	10.469	7	300	161	−59	11.932	162	3½	75½°N	171½°W
8	Gabbro	43°58′	71°28′	12	0	3 36	9.91	19	−55½	11.638	30	150	165½	−49½	11.790	52	6	72°N	153½°E
9	Monzonite	43° 7′	71°12′	12	0	1.19	0.32	59	−67½	1.891*	1	250	352	+58½	11.823	62	5½	83°N	168½°E
10	Monzonite	43° 7′	71°12′	12	0	1.34	0 53	355	+37½	9 511	4	150	351	+58	11.927	150	1	82°N	168½°E
11	Essexite	44°55′	71°31′	8	2	6.48	5.67	351	+55½	7 761	29	150	1	+65½	7.948	135	5	87½°N	55°W
12	Lamprophyre	44°55′	71°31′	10	0	1.35	0.65	19	+78	9.632	24	250	5	+63	9.985	581	2	86½°N	25°E
A	Gabbro	43°27′	72°30′	12	0	5.84	2.66	9	+73	11.326	16	250	153	−52	11.971	384	2	66°N	179½°W
B	Gabbro	43°27′	72°30′	12	0	5.32	2.05	35½	+86½	9.919	5	250	146	−55	11.964	308	2½	62½°N	167½°W

Notes:

K is the susceptibility in emu/cm³.

J is the NRM intensity in emu/cm³ $\times 10^{-3}$.

M_0 gives the NRM results, and M_{ac} gives those after demagnetization in a peak alternating field of B_{ac} oe.

N is the number of samples included in the analysis and E is the number excluded.

R is the length of the resultant of the N unit vectors and D and I are the declination and inclination of its direction.

α_{95} is the semiangle of the cone of 95% confidence, and $k = (N - 1)/(N - R)$ is the best estimate of Fisher's dispersion parameter.

* The resultant R is less than the value R_0 tabulated by *Watson* [1956], which is exceeded in only 5% of the cases by the resultant of N randomly directed vectors.

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Alternating Field Demagnetization

All samples were demagnetized in an alternating field with equipment that is in principle much like that described by *Creer* [1959] and *McElhinny* [1966]. Two specimens from a single core from each site were progressively demagnetized in steps until a stable direction of magnetization was reached. The remaining specimens from the site were then demagnetized in the field which had been thus determined. For those sites where the direction of magnetization in the specimens continued to change on application of higher fields, several more specimens were progressively demagnetized to determine the peak alternating field at which the dispersion was least.

The NRM results of samples from sites 6, 7, and 8 collected at Slide Brook on Mount Tripyramid gave directions reversed with respect to the earth's present field. When specimens from these sites were treated in alternating fields of up to 700 oe, the mean inclination remained negative, but there was a significant change in declination with an increase in the value of *Fisher's* [1953] precision parameter k. For site 7 the value of k increased from 7 to 162. The demagnetization curves for samples from these sites are typical of those for stable or partially stable rocks, and at peak fields of 250 oe the intensity is still 30% of the original NRM (curve 1, Figure 2). Most of the directions of NRM for site 9 in the monzonite of Mount Pawtuckaway are reversely magnetized. By progressive demagnetization they very quickly become negatively (normally) magnetized at peak fields as low as 50 oe. The magnetic moment of these specimens also drops off very quickly (curve 3, Figure 2). Only 20% of the intensity of NRM remains in peak fields of 50 oe. On application of higher fields the intensity and direction change very little, so that at a peak value of 500 oe the intensity is still 12% of the NRM intensity. It is probable that these specimens acquired a viscous component while being stored in the laboratory. The NRM grouping for this site is not statistically significant. After the viscous component has been removed by partial alternating field demagnetization, the grouping is statistically significant with a k of 62.

For most sites the unstable component was

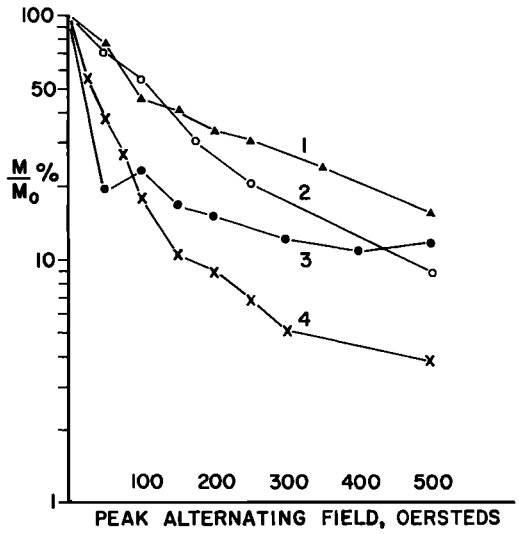


Fig. 2. Normalized demagnetization curves of specimens from 5 sites. Curve 1 is from site 6, curve 2 is from site 11, curve 3 is from site 9, and curve 4 is from site 5. M/M_0 is the ratio of remaining to initial intensity after treatment in the corresponding field. The plot is semilogarithmic, the abscissa being the log scale.

completely removed in peak fields of 150 or 250 oe. However, for the three gabbro sites of the Merrymeeting stock and the monzonite of the Belknap Mountains, higher fields had to be applied. Most sites give well-grouped directions with either positive or negative polarities. Table 1 shows the marked improvement in grouping for most of the sites after ac demagnetization. Figure 3 shows the mean directions of magnetization for the sites after cleaning. We were unable to isolate any stable component from the diorites. The diorite of the Belknap Mountains is very inhomogeneous and contains many inclusions. The diorite at Mount Pawtuckaway is foliated, and the rock is believed to be anisotropic. Since no concordant information was obtained from these sites, they will not be considered further.

Discussion of Results

After partial alternating field demagnetization, the site mean directions fall into two groups. One group which comprises both the monzonite from the Belknap and Pawtuckaway mountains and the essexite and lamprophyre from Mount Monadnock show northward-inclined normal directions with negative polari-

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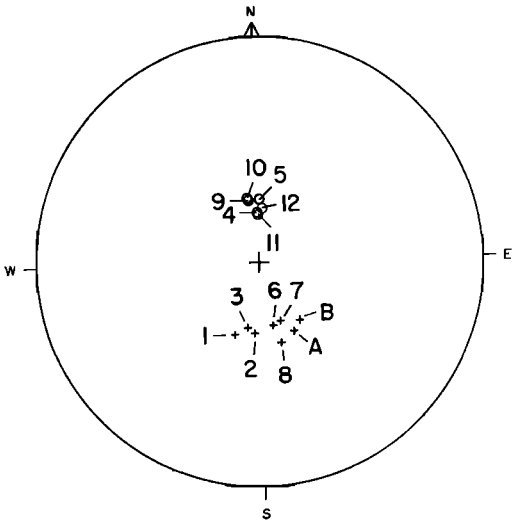


Fig. 3. Stereographic projection of the site mean directions. The open circles are north-seeking directions down and the crosses are north-seeking directions up.

ties. The other group, which contains gabbros from the Merrymeeting stock, Mount Tripyramid, and Mount Ascutney, have north-seeking directions upward to the south. The directions of the latter group are reversed with respect to the first group. In Figure 3 it can be seen that the direction of magnetization for sites A and B at Mount Ascutney, which has been dated at 125 to 135 m.y. [*Faul et al.*, 1963], is not very different from that for the other sites, in particular sites 6, 7, and 8 at Mount Tripyramid. The pole positions from rocks collected at Mount Ascutney agree very well with those from dated Upper Jurassic and Cretaceous rocks of the North American continent (Figures 4 and 5) [*Larochelle*, 1962; *Grommé and Merrill*, 1965*a*, *b*].

On paleomagnetic grounds alone there is insufficient reason to separate these sites from the sites in the White Mountain series. However, these two sites have been excluded from the White Mountain series in calculating a mean paleomagnetic pole position because of the young age given by K-Ar. The fact that the pole position of the samples from Mount Tripyramid lies close to the pole position of those from Mount Ascutney (Figure 4) might suggest that this ring dike is younger than the larger instrusions of the White Mountain series. Some support for this contention is provided by the fact that the Mount Tripyramid ring dike is apparently intruded into the Conway granite. The latter is supposed to be the youngest rock type in the White Mountain series. Because the evidence at the present time is only suggestive, the results for the sites at Mount Tripyramid have been combined with the others in determining a mean pole position, and the results for sites A and B have been excluded because of their independently determined lesser age.

The paleomagnetic pole position for the White Mountain plutonic-volcanic series, calculated from the mean directions of magnetization, is at 85½°N and 124½°E. The parameters are: 12 sites; declination (D) = 358½°; inclination (I) = +59°; resultant (R) = 11.900; precision of site poles (K) = 110; the semiaxes of the oval of 95% confidence for the mean pole positions p and m = 4½° and 6°, respectively; and the semiangle of the cone of 95% confidence α_{95} = 4. The pole position calculated by combining the site mean poles is at 85½°N and 126½°E. The parameters are: 12 sites, R = 11.828, K = 64, and α_{95} = 5½.

It can be seen in Figure 5 that in relation to the other Mesozoic pole positions the White Mountain pole falls surprisingly near the pres-

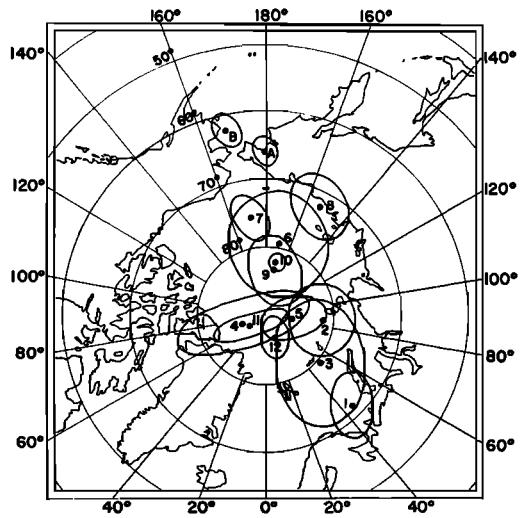


Fig. 4. Paleomagnetic pole position and 95% ovals of confidence calculated from the directions at each site. The plot is on a stereographic projection of the north polar region. The numbers and letters correspond to the site designations in Table 1.

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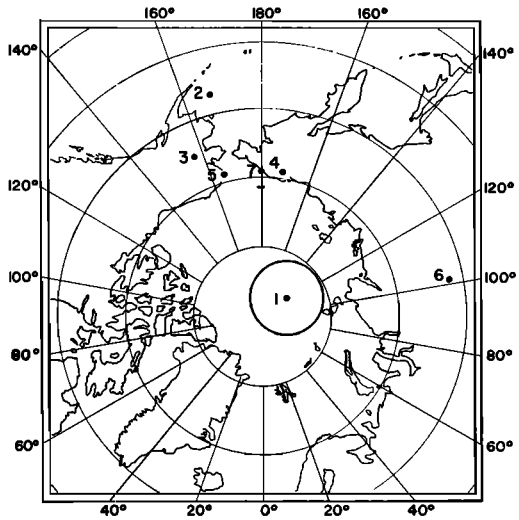


Fig. 5. (1) Mean paleomagnetic pole position for the White Mountain plutonic-volcanic series with its 95% circle of confidence. (2) The paleomagnetic pole from the Upper Jurassic Valley series batholiths of California [*Grommé and Merrill,* 1965*b*]. The Cretaceous poles from (3) the Monteregian Hills, (4) Mount Megantic in Quebec [*Larochelle,* 1962], (5) the Sierra Nevada granites [*Grommé and Merrill,* 1965*a*], and (7) the Isachsen diabase [*Larochelle and Black,* 1963]. (6) The mean Triassic pole position for North America [*Beck,* 1965].

ent north pole. The difference in the age of the White Mountain plutonic-volcanic series, which is placed at 180 m.y., and that of the Upper Triassic of New Jersey, which has been determined by *Erickson and Kulp* [1960] to be 190 m.y., is small; yet the difference in directions of magnetization is great. For instance, the mean inclination determined from the New Jersey Triassic [*Opdyke,* 1961] is about 25°, and the mean inclination of the White Mountain series is 59°; in declination, however, they are almost identical. The variation between the Lower Jurassic White Mountain series and the Upper Jurassic and Cretaceous directions is largely a difference in declination. It is clear that the observed difference in inclination is not attributable to viscous magnetization in the rocks because they have all been magnetically cleaned and also because a number of the sites show reversed polarities.

Data from the Jurassic and Cretaceous periods of the northern hemisphere are rather few. The mid-Mesozoic paleomagnetic record is more complete in both Africa [*Gough et al.,* 1964] and Australia [*Irving et al.,* 1963]. The paleomagnetic record from these southern continents does not show any large shift in pole position during Early Jurassic time. It is hoped that further paleomagnetic research in North America and Europe will lead to a better understanding of the magnetic record during this period of time.

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